



Software Architectures

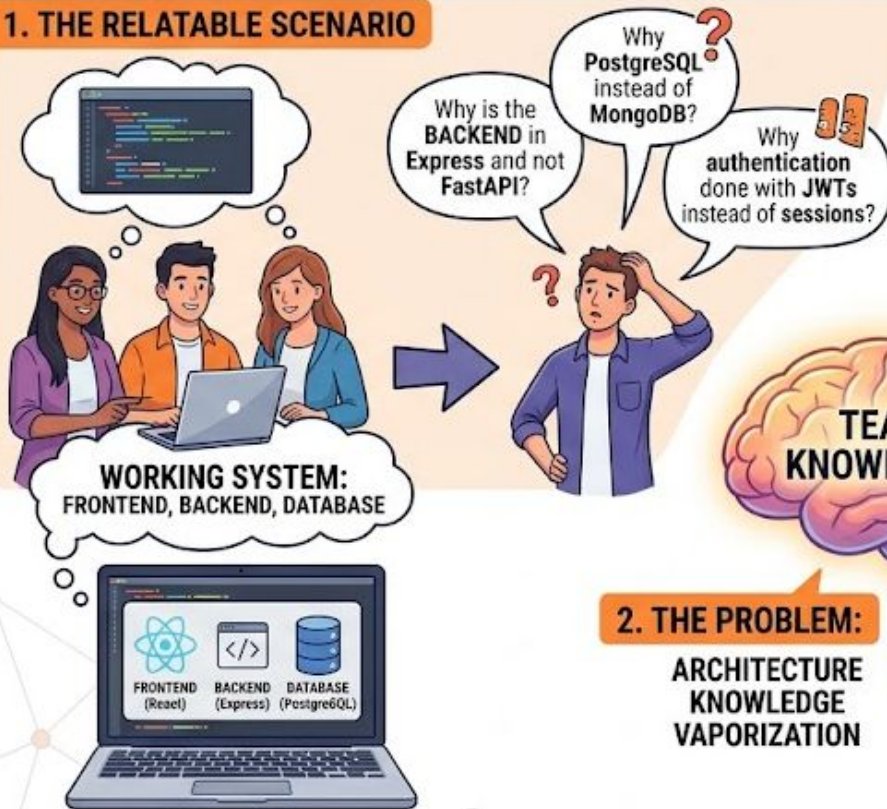
Practice 10

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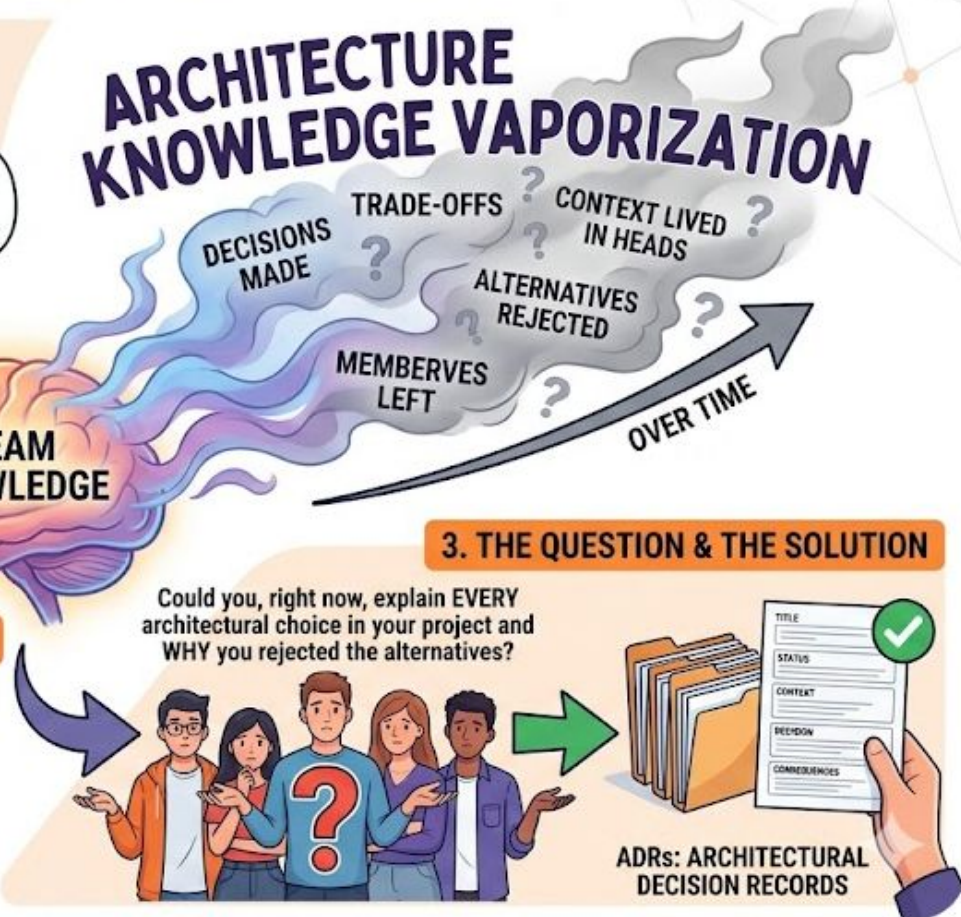
Architectural Decisions

The Problem of Lost Decisions

1. THE RELATABLE SCENARIO



2. THE PROBLEM: ARCHITECTURE KNOWLEDGE VAPORIZATION



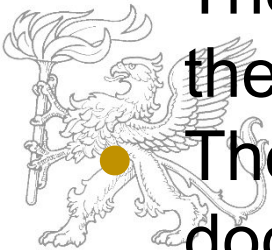
3. THE QUESTION & THE SOLUTION

Could you, right now, explain EVERY architectural choice in your project and WHY you rejected the alternatives?



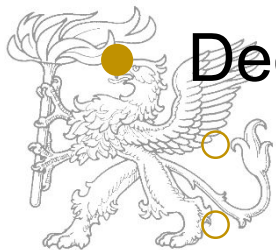
Architecture Decision Records (ADRs)

- An ADR is not a design document or a spec. It captures a single decision, not an entire design.
- ADRs are immutable — once accepted, you don't edit them. If you reverse a decision, you write a new ADR that supersedes the old one. This preserves the historical trail.
- They are lightweight — typically one page or less. The goal is low friction so people actually write them.
- They live alongside the code, usually in a docs/adr/ or docs/decisions/ folder in the repository. They are versioned with the codebase.



What can and what cannot be an ADR?

- Decisions that are hard or costly to reverse 
 - Database technology
 - Frontend framework
 - Auth strategy
 - API style
 - State management approach
 - Error taxonomy
 - Deployment model

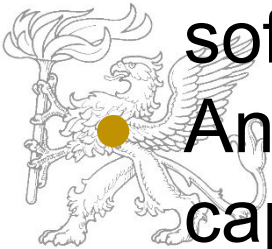


● Decisions that are easily reversible 

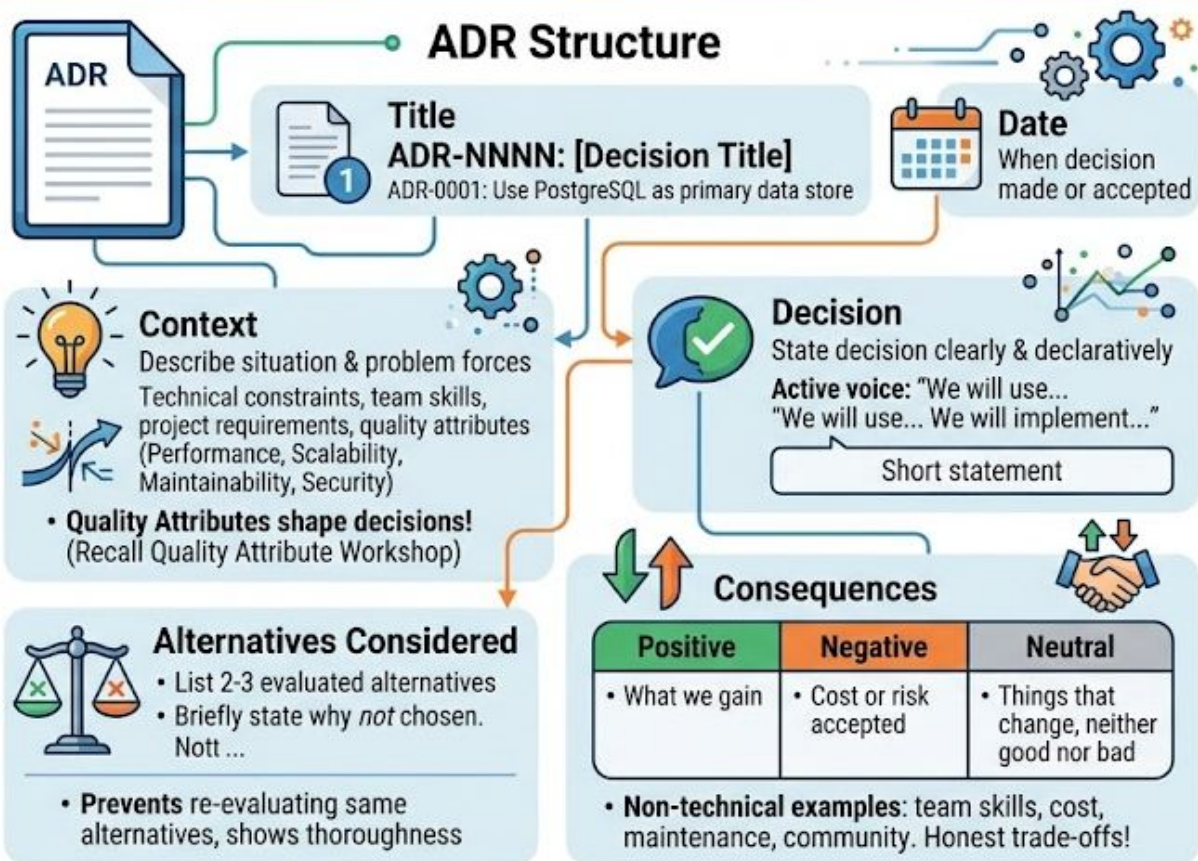
- Variable naming conventions
- Which CSS color to use for buttons
- Choosing between 2 packages that has the same goal

Definitions

- **Architectural Decision (AD)** is a justified design choice that addresses a functional or non-functional requirement that is architecturally significant.
- An **Architecturally Significant Requirement (ASR)** is a requirement that has a measurable effect on the architecture and quality of a software and/or hardware system.
- An **Architectural Decision Record (ADR)** captures a single AD and its rationale.



ADR: Architecture Decision Record Structure & Lifecycle



ADR Status Lifecycle



Proposed: Under consideration

Accepted: Official decision

Deprecated: No longer recommended

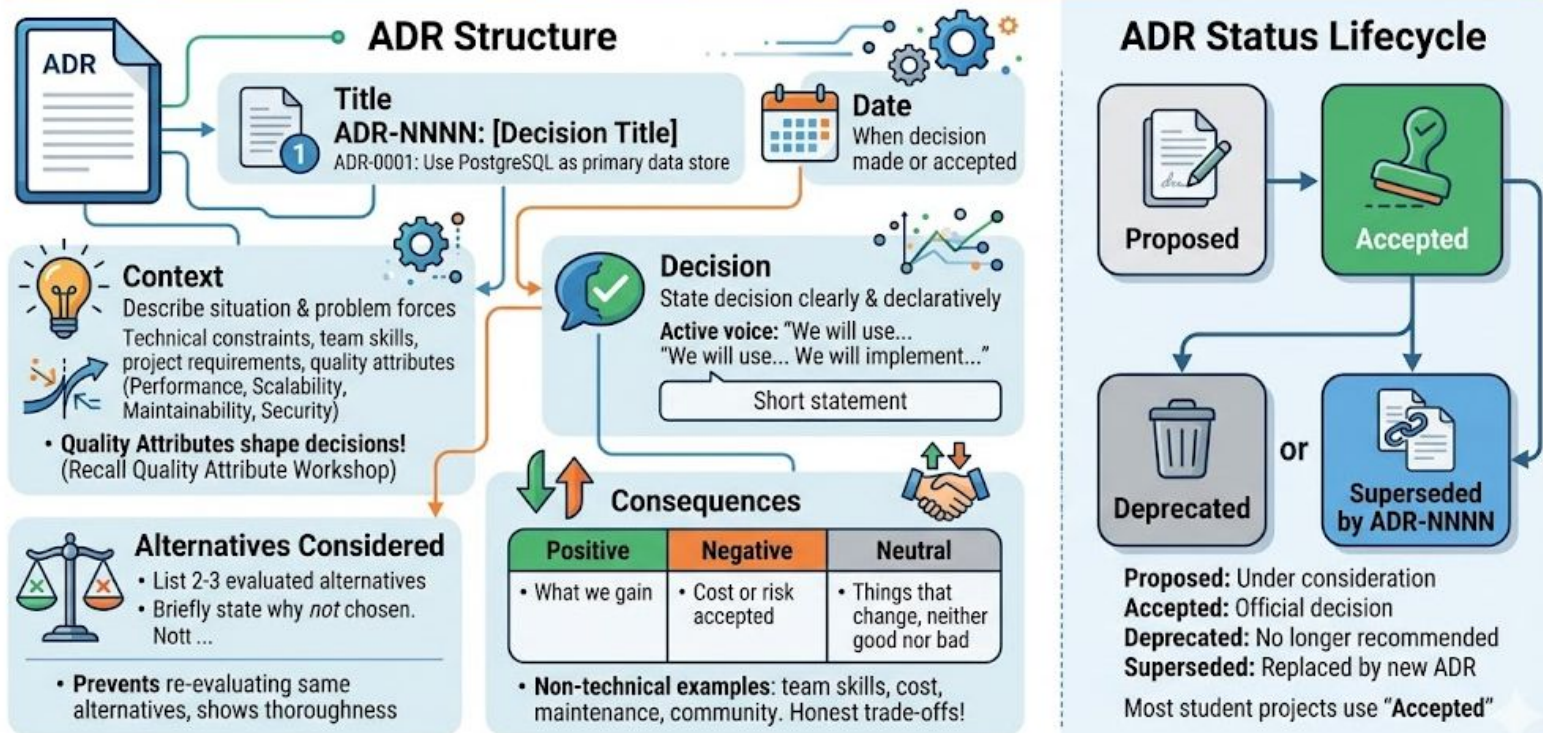
Superseded: Replaced by new ADR

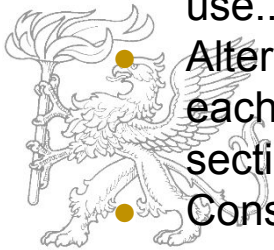
Most student projects use "Accepted"

The Standard ADR template

- Multiple templates are available ([LINK](#))
- A template proposed by Michael Nygard ([LINK](#))

ADR: Architecture Decision Record Structure & Lifecycle





Michael Nygard's template

- Title: A short noun phrase with a sequential number.
Format: ADR-NNNN: <Decision Title>.
Example: ADR-0001: Use PostgreSQL as the primary data store.
- Status: One of Proposed, Accepted, Deprecated, Superseded by ADR-NNNN.
- Date: When the decision was made (or accepted).
- Context: Describe the situation and the problem that needs a decision. What forces are at play? What constraints exist? This is the "why are we even talking about this" section.
- Emphasize: this should include technical constraints, team constraints, project requirements, and any relevant quality attributes (performance, scalability, maintainability, security).
- Decision: State the decision clearly and declaratively. Use active voice: "We will use...", "We will implement...".
- Alternatives Considered: List 2–3 alternatives that were seriously evaluated. For each, briefly state why it was not chosen. This is arguably the most valuable section.
- Consequences: What follows from this decision? Include both positive and negative consequences. Be honest — every decision has trade-offs.
 - Positive (what we gain), negative (what we accept as a cost or risk), neutral (things that change but are neither good nor bad)

Common mistakes and anti-patterns

COMMON ADR ANTIPATTERNS TO AVOID: WHAT **NOT** TO DO.

POPULARITY ≠ JUSTIFICATION.



POPULARITY ≠ JUSTIFICATION.

- Popularity is not architectural reasoning.
- Explain why the popular choice fits YOUR specific constraints and quality attributes.

NO REAL ALTERNATIVES (CHECKING THE BOX).



NO REAL ALTERNATIVES (CHECKING THE BOX).

- Listing one terrible option just to 'check the box' doesn't count.
- Alternatives should be genuinely plausible choices.

TOO VAGUE. "WE USE A MODERN FRAMEWORK"



TOO VAGUE. "WE USE A MODERN FRAMEWORK" SAYS NOTHING.

- Be specific about what, why, and what it implies.
- Vagueness hinders understanding of the decision.

WRITING ADRs AFTER THE FACT WITHOUT HONESTY.



WRITING ADRs AFTER THE FACT WITHOUT OBJECTIVE HONESTY.

- If you chose it due to team familiarity, own that valid reason.
- Transparency is better than pretending you did an objective evaluation.

FORGETTING NEGATIVE CONSEQUENCES.



FORGETTING NEGATIVE CONSEQUENCES

- Every decision has downsides and trade-offs.
- If the consequences are all positive, something is missing.